

P-SV Wave Reflection and Transmission (R/T) Coefficients

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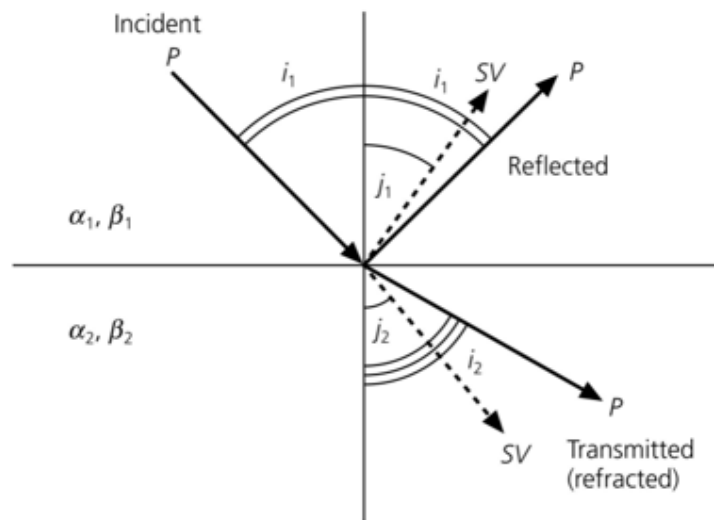


Figure 1: P-SV R/T Coefficients

This note covers **P-SV wave reflection and transmission (R/T) coefficients**, from the full Zoeppritz equations to their linearized form (Shuey's equation) and the special case of the free surface. The P-SV problem is significantly more complex than the SH-case because P-waves and SV-waves couple at an interface. When a P-wave is incident on an interface between two elastic media, it generates four waves (1):

- **Reflected P-wave** (travels back in medium 1)
- **Reflected SV-wave** (mode conversion in medium 1)
- **Transmitted P-wave** (refracted into medium 2)
- **Transmitted SV-wave** (mode conversion in medium 2)

This is fundamentally different from SH waves, where no mode conversion occurs because SH motion is perpendicular to the plane of incidence. This leads to a system of four equations instead of two.

1. Concept Overview of P-SV Wave Propagation

Problem Setup

We consider an interface between two elastic solids:

- **Medium 1:** P-wave velocity α_1 , S-wave velocity β_1 , density ρ_1 , Lamé parameters λ_1, μ_1
- **Medium 2:** P-wave velocity α_2 , S-wave velocity β_2 , density ρ_2 , Lamé parameters λ_2, μ_2

Snell's Law

For an incident plane wave (P or SV), Snell's Law governs the directions of all generated waves:

$$\frac{\sin i_1}{\alpha_1} = \frac{\sin j_1}{\beta_1} = \frac{\sin i_2}{\alpha_2} = \frac{\sin j_2}{\beta_2} = p$$

where p is the ray parameter, constant for all waves; i_1 is the angle of the incident (and, by symmetry, the reflected) P-wave, j_1 that of the reflected SV-wave, and i_2, j_2 those of the transmitted P- and SV-waves (Figure 1). All angles are measured from the interface normal. Equivalently, $k_x = \omega p$ is the horizontal wavenumber shared by all five waves.

Potential Functions

To handle the coupled P-SV system, we use scalar potential ϕ for P-waves and vector potential ψ for SV-waves, where the displacement $\mathbf{u}$ is:

$$\mathbf{u} = \nabla\phi + \nabla \times \psi$$

P-waves (dilatational):

$$\mathbf{u}_P = \nabla\phi \tag{1}$$

SV-waves (shear, vertical polarization):

$$\mathbf{u}_{SV} = \nabla \times (0, \psi, 0) \tag{2}$$

For 2D P-SV case in x-z plane, the vector potential has only a y-component: $\psi = (0, \psi, 0)$.

Boundary Conditions

At interface $z = 0$, we require continuity of:

1. **Displacement:** u_x and u_z

$$\begin{pmatrix} u_x \\ u_z \end{pmatrix} = \begin{pmatrix} \partial_x & -\partial_z \\ \partial_z & \partial_x \end{pmatrix} \begin{pmatrix} \phi \\ \psi \end{pmatrix} \quad (3)$$

This leads to the displacement components:

$$u_x = \frac{\partial \phi}{\partial x} - \frac{\partial \psi}{\partial z}, \quad u_z = \frac{\partial \phi}{\partial z} + \frac{\partial \psi}{\partial x}$$

2. **Stress:** σ_{zz} (normal stress) and σ_{xz} (shear stress)

Stress expressions in terms of potentials:

Remember:

$$\sigma_{ij} = \lambda \delta_{ij} \epsilon_{kk} + 2\mu \epsilon_{ij}$$

$$\epsilon_{kk} = \nabla \cdot u = \epsilon_{11} + \epsilon_{22} + \epsilon_{33}$$

$$\sigma_{zz} = \lambda \nabla^2 \phi + 2\mu \left(\frac{\partial^2 \phi}{\partial z^2} + \frac{\partial^2 \psi}{\partial x \partial z} \right)$$

$$\sigma_{xz} = \mu \left(2 \frac{\partial^2 \phi}{\partial x \partial z} + \frac{\partial^2 \psi}{\partial x^2} - \frac{\partial^2 \psi}{\partial z^2} \right)$$

The Zoeppritz Equations:

Applying these four boundary conditions to the complete wavefield (incident P, reflected P, reflected SV, transmitted P, transmitted SV) results in a 4x4 system of linear equations. This full system is the Zoeppritz equations. Solving them gives the exact R/T coefficients for a planar interface between two solids. They are the "ground truth" but are algebraically complex and non-intuitive.

2. Wave Equations in Each Medium

Medium 1:

$$\phi_1 = A_1 e^{i(\omega t - k_x x + k_z \alpha_1 z)} + A' e^{i(\omega t - k_x x - k_z \alpha_1 z)} \quad (4)$$

$$\psi_1 = B_1 e^{i(\omega t - k_x x - k_z \beta_1 z)} \quad (5)$$

Medium 2:

$$\phi_2 = A_2 e^{i(\omega t - k_x x + k_z \alpha_2 z)} \quad (6)$$

$$\psi_2 = B_2 e^{i(\omega t - k_x x + k_z \beta_2 z)} \quad (7)$$

Sign conventions (time convention $e^{i\omega t}$, with $k_z > 0$ real below the critical angles):

- A term $e^{i(\omega t - k_x x + k_z z)}$ propagates in the $-z$ direction — from medium 1 toward medium 2 (the incident and transmitted waves: A_1, A_2, B_2).
- A term $e^{i(\omega t - k_x x - k_z z)}$ propagates in the $+z$ direction — back into medium 1 (the reflected waves: A', B_1).

Medium 1 therefore occupies $z > 0$ and medium 2 occupies $z < 0$. Taking z positive upward, this is the picture of Figure 1 (downgoing incident wave in the upper medium); taking z positive downward, it is the mirror image used in the free-surface geometry of Figure 2 (upgoing incident wave in the halfspace below the surface). The algebra is identical in both readings.

Wavenumber Relations

From Snell's law, k_x is continuous across the interface:

$$k_{z\alpha_m} = \sqrt{\frac{\omega^2}{\alpha_m^2} - k_x^2}, \quad k_{z\beta_m} = \sqrt{\frac{\omega^2}{\beta_m^2} - k_x^2}, \quad m = 1, 2 \quad (8)$$

Beyond a critical angle the corresponding radicand becomes negative and k_z becomes imaginary; the branch must then be chosen so that the wave decays away from the interface (evanescent/inhomogeneous wave).

3. Displacement Fields

Computing Displacements at Interface from Potentials

To handle the coupled P-SV system, we use scalar potential ϕ for P-waves and vector potential ψ for SV-waves, where the displacement $\mathbf{u}$ is:

$$\mathbf{u} = \nabla \phi + \nabla \times \psi$$

P-waves (dilatational):

P-waves are **irrotational** (no rotation) and can be described by a scalar potential ϕ such that:

$$\mathbf{u}_P = \nabla \phi \quad (9)$$

In component form:

$$\boxed{u_x = \frac{\partial \phi}{\partial x}, \quad u_z = \frac{\partial \phi}{\partial z}} \quad (10)$$

No u_y component because we are considering 2D.

Take Note:

Why this works

Wave equation for P-waves

$$\nabla^2 \phi = \frac{1}{\alpha^2} \frac{\partial^2 \phi}{\partial t^2}$$

Checking irrotationality

$$\nabla \times \mathbf{u}_P = \nabla \times (\nabla \phi) = 0$$

This is always true because the curl of a gradient is identically zero.

Dilatation (volume change)

$$\nabla \cdot \mathbf{u}_P = \nabla \cdot (\nabla \phi) = \nabla^2 \phi \neq 0$$

SV-waves (shear, vertical polarization):

SV-waves are **solenoidal** (divergence-free) and can be described by a vector potential. For 2D motion in the x-z plane, we use:

$$\mathbf{u}_{SV} = \nabla \times (0, \psi, 0) = \nabla \times (\psi \hat{y}) \quad (11)$$

For 2D P-SV case in x-z plane, the vector potential has only a y-component: $\psi = (0, \psi, 0)$.

Detailed derivation

The curl in Cartesian coordinates is:

$$\nabla \times \mathbf{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \quad (12)$$

For $\mathbf{A} = (0, \psi, 0)$:

$$\nabla \times (0, \psi, 0) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & \psi & 0 \end{vmatrix} \quad (13)$$

Computing each component:

x-component:

$$(\nabla \times \mathbf{A})_x = \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} = \frac{\partial 0}{\partial y} - \frac{\partial \psi}{\partial z} = -\frac{\partial \psi}{\partial z} \quad (14)$$

For 2D problems where nothing depends on y ($\partial/\partial y = 0$):

Standard curl calculation:

$$(\nabla \times \mathbf{A})_x = \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} = -\frac{\partial \psi}{\partial z} \quad (15)$$

$$(\nabla \times \mathbf{A})_y = \frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} = 0 \quad (16)$$

$$(\nabla \times \mathbf{A})_z = \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} = \frac{\partial \psi}{\partial x} \quad (17)$$

$$u_x = -\frac{\partial \psi}{\partial z}, \quad u_z = \frac{\partial \psi}{\partial x} \quad (18)$$

So:

$$\nabla \times \boldsymbol{\psi} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & \psi & 0 \end{vmatrix} = -\hat{x} \frac{\partial \psi}{\partial z} + \hat{z} \frac{\partial \psi}{\partial x} \quad (19)$$

Therefore:

$$\boxed{u_x = -\frac{\partial \psi}{\partial z}, \quad u_z = +\frac{\partial \psi}{\partial x}} \quad (20)$$

Why this works

Wave equation for SV-waves

$$\nabla^2 \psi = \frac{1}{\beta^2} \frac{\partial^2 \psi}{\partial t^2} \quad (21)$$

Checking solenoidality (divergence-free)

$$\nabla \cdot \mathbf{u}_{SV} = \frac{\partial u_x}{\partial x} + \frac{\partial u_z}{\partial z} = \frac{\partial^2 \psi}{\partial x \partial z} - \frac{\partial^2 \psi}{\partial z \partial x} = 0 \quad (22)$$

SV-waves involve no volume change.

Rotation (vorticity)

$$\nabla \times \mathbf{u}_{SV} = \nabla \times (\nabla \times (0, \psi, 0)) \neq 0 \quad (23)$$

SV-waves involve rotation/shear.

So, from equations 10 and 20:

Horizontal displacement:

$$u_x = \frac{\partial \phi}{\partial x} - \frac{\partial \psi}{\partial z} \quad (24)$$

Vertical displacement:

$$u_z = \frac{\partial \phi}{\partial z} + \frac{\partial \psi}{\partial x} \quad (25)$$

In matrix form, the continuity of displacement at the interface becomes:

$$\begin{pmatrix} u_x \\ u_z \end{pmatrix} = \begin{pmatrix} \partial_x & -\partial_z \\ \partial_z & \partial_x \end{pmatrix} \begin{pmatrix} \phi \\ \psi \end{pmatrix} \quad (26)$$

4. Computing Stress Components in Terms of Potentials

General Stress-Strain Relations

For an isotropic elastic medium, the stress tensor is:

$$\sigma_{ij} = \lambda \delta_{ij} \epsilon_{kk} + 2\mu \epsilon_{ij} \quad (27)$$

where:

- λ, μ are Lamé parameters

- δ_{ij} is the Kronecker delta
- $\epsilon_{kk} = \epsilon_{11} + \epsilon_{22} + \epsilon_{33}$ is the dilatation (volumetric strain)
- $\epsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$ is the strain tensor

Step 1: Express Dilatation in Terms of Displacement

The dilatation is:

$$\epsilon_{kk} = \nabla \cdot \mathbf{u} = \frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z} \quad (28)$$

For 2D problems in the x-z plane (no y-dependence):

$$\epsilon_{kk} = \frac{\partial u_x}{\partial x} + \frac{\partial u_z}{\partial z} \quad (29)$$

Step 2: Express Displacements in Terms of Potentials

Recall:

- P-wave: $\mathbf{u}_P = \nabla \phi$, so $u_x = \frac{\partial \phi}{\partial x}$, $u_z = \frac{\partial \phi}{\partial z}$
- SV-wave: $\mathbf{u}_{SV} = \nabla \times (0, \psi, 0)$, so $u_x = -\frac{\partial \psi}{\partial z}$, $u_z = \frac{\partial \psi}{\partial x}$

Total displacement:

$$u_x = \frac{\partial \phi}{\partial x} - \frac{\partial \psi}{\partial z} \quad (30)$$

$$u_z = \frac{\partial \phi}{\partial z} + \frac{\partial \psi}{\partial x} \quad (31)$$

Step 3: Calculate Dilatation

$$\epsilon_{kk} = \frac{\partial u_x}{\partial x} + \frac{\partial u_z}{\partial z} \quad (32)$$

$$= \frac{\partial}{\partial x} \left(\frac{\partial \phi}{\partial x} - \frac{\partial \psi}{\partial z} \right) + \frac{\partial}{\partial z} \left(\frac{\partial \phi}{\partial z} + \frac{\partial \psi}{\partial x} \right) \quad (33)$$

$$= \frac{\partial^2 \phi}{\partial x^2} - \frac{\partial^2 \psi}{\partial x \partial z} + \frac{\partial^2 \phi}{\partial z^2} + \frac{\partial^2 \psi}{\partial z \partial x} \quad (34)$$

$$= \frac{\partial^2 \phi}{\partial x^2} - \cancel{\frac{\partial^2 \psi}{\partial x \partial z}} + \frac{\partial^2 \phi}{\partial z^2} + \cancel{\frac{\partial^2 \psi}{\partial z \partial x}} \quad (35)$$

$$\epsilon_{kk} = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial z^2} = \nabla^2 \phi \quad (36)$$

So:

$$\boxed{\epsilon_{kk} = \nabla \cdot \mathbf{u} = \nabla^2 \phi} \quad (37)$$

This shows that the dilatation depends only on the P-wave potential ϕ , not on the SV-wave potential ψ .

Step 4: Derive σ_{zz}

From the general formula (equation 27):

$$\sigma_{zz} = \lambda \epsilon_{kk} + 2\mu \epsilon_{zz} \quad (38)$$

where:

$$\epsilon_{zz} = \frac{\partial u_z}{\partial z} \quad (39)$$

Computing ϵ_{zz}

$$\epsilon_{zz} = \frac{\partial u_z}{\partial z} = \frac{\partial}{\partial z} \left(\frac{\partial \phi}{\partial z} + \frac{\partial \psi}{\partial x} \right) = \frac{\partial^2 \phi}{\partial z^2} + \frac{\partial^2 \psi}{\partial x \partial z} \quad (40)$$

Substituting into σ_{zz}

$$\sigma_{zz} = \lambda \nabla^2 \phi + 2\mu \left(\frac{\partial^2 \phi}{\partial z^2} + \frac{\partial^2 \psi}{\partial x \partial z} \right) \quad (41)$$

$$= \lambda \left(\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial z^2} \right) + 2\mu \frac{\partial^2 \phi}{\partial z^2} + 2\mu \frac{\partial^2 \psi}{\partial x \partial z} \quad (42)$$

$$= \lambda \frac{\partial^2 \phi}{\partial x^2} + \lambda \frac{\partial^2 \phi}{\partial z^2} + 2\mu \frac{\partial^2 \phi}{\partial z^2} + 2\mu \frac{\partial^2 \psi}{\partial x \partial z} \quad (43)$$

$$= \lambda \left(\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial z^2} \right) + 2\mu \frac{\partial^2 \phi}{\partial z^2} + 2\mu \frac{\partial^2 \psi}{\partial x \partial z} \quad (44)$$

Note that $\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial z^2}$, So, this can be rewritten as:

$$\sigma_{zz} = \lambda \nabla^2 \phi + 2\mu \frac{\partial^2 \phi}{\partial z^2} + 2\mu \frac{\partial^2 \psi}{\partial x \partial z} \quad (45)$$

$$\boxed{\sigma_{zz} = \lambda \nabla^2 \phi + 2\mu \left(\frac{\partial^2 \phi}{\partial z^2} + \frac{\partial^2 \psi}{\partial x \partial z} \right)} \quad (46)$$

Step 5: Derive σ_{xz}

The shear stress is:

$$\sigma_{xz} = 2\mu\epsilon_{xz} = 2 \cdot \mu \cdot \frac{1}{2} \left(\frac{\partial u_x}{\partial z} + \frac{\partial u_z}{\partial x} \right) = \mu \left(\frac{\partial u_x}{\partial z} + \frac{\partial u_z}{\partial x} \right) \quad (47)$$

Computing Each Term

First term:

$$\frac{\partial u_x}{\partial z} = \frac{\partial}{\partial z} \left(\frac{\partial \phi}{\partial x} - \frac{\partial \psi}{\partial z} \right) = \frac{\partial^2 \phi}{\partial x \partial z} - \frac{\partial^2 \psi}{\partial z^2} \quad (48)$$

Second term:

$$\frac{\partial u_z}{\partial x} = \frac{\partial}{\partial x} \left(\frac{\partial \phi}{\partial z} + \frac{\partial \psi}{\partial x} \right) = \frac{\partial^2 \phi}{\partial z \partial x} + \frac{\partial^2 \psi}{\partial x^2} \quad (49)$$

Adding the Terms

$$\frac{\partial u_x}{\partial z} + \frac{\partial u_z}{\partial x} = \frac{\partial^2 \phi}{\partial x \partial z} - \frac{\partial^2 \psi}{\partial z^2} + \frac{\partial^2 \phi}{\partial x \partial z} + \frac{\partial^2 \psi}{\partial x^2} \quad (50)$$

$$= 2 \frac{\partial^2 \phi}{\partial x \partial z} - \frac{\partial^2 \psi}{\partial z^2} + \frac{\partial^2 \psi}{\partial x^2} \quad (51)$$

Final Expression for σ_{xz}

$$\sigma_{xz} = \mu \left(2 \frac{\partial^2 \phi}{\partial x \partial z} - \frac{\partial^2 \psi}{\partial z^2} + \frac{\partial^2 \psi}{\partial x^2} \right) \quad (52)$$

$$\boxed{\sigma_{xz} = \mu \left(2 \frac{\partial^2 \phi}{\partial x \partial z} + \frac{\partial^2 \psi}{\partial x^2} - \frac{\partial^2 \psi}{\partial z^2} \right)} \quad (53)$$

The fundamental relationships are:

$$\boxed{\epsilon_{kk} = \nabla \cdot \mathbf{u} = \epsilon_{11} + \epsilon_{22} + \epsilon_{33} = \nabla^2 \phi} \quad (54)$$

$$\boxed{\sigma_{zz} = \lambda \nabla^2 \phi + 2\mu \left(\frac{\partial^2 \phi}{\partial z^2} + \frac{\partial^2 \psi}{\partial x \partial z} \right) = \lambda (\partial_{xx} \phi + \partial_{zz} \phi) + 2\mu (\partial_{zz} \phi + \partial_{xz} \psi)} \quad (55)$$

$$\boxed{\sigma_{xz} = \mu \left(2 \frac{\partial^2 \phi}{\partial x \partial z} + \frac{\partial^2 \psi}{\partial x^2} - \frac{\partial^2 \psi}{\partial z^2} \right) = \mu (2\partial_{xz} \phi + \partial_{xx} \psi - \partial_{zz} \psi)} \quad (56)$$

Physical Interpretation

- The dilatation ϵ_{kk} depends only on the P-wave potential ϕ , confirming that only P-waves cause volume changes.

- The normal stress σ_{zz} involves both ϕ and ψ , showing coupling between P and SV waves at boundaries.
- The shear stress σ_{xz} also involves both potentials, demonstrating mode conversion.
- These expressions allow us to write boundary conditions entirely in terms of ϕ and ψ , which is essential for deriving the Zoeppritz equations.

5. Boundary Conditions at $z=0$

At an interface (say $z = 0$), we require:

Four Boundary Conditions:

1. Continuity of horizontal displacement: $u_x^{(1)} = u_x^{(2)}$
2. Continuity of vertical displacement: $u_z^{(1)} = u_z^{(2)}$
3. Continuity of shear stress: $\sigma_{xz}^{(1)} = \sigma_{xz}^{(2)}$
4. Continuity of normal stress: $\sigma_{zz}^{(1)} = \sigma_{zz}^{(2)}$

At a free surface ($z = 0$), we require:

1. $\sigma_{xz} = 0$
2. $\sigma_{zz} = 0$

These conditions, expressed in terms of ϕ and ψ using the formulas derived above, lead to the famous Zoeppritz equations for reflection and transmission coefficients. These four equations also determine the four unknowns: A' , B_1 , A_2 , B_2 .

6. Detailed Derivation of Boundary Conditions

Computing Derivatives

For $\phi_1 = A_1 e^{i(\omega t - k_x x + k_{z\alpha_1} z)} + A' e^{i(\omega t - k_x x - k_{z\alpha_1} z)}$ at $z = 0$:

$$\left. \frac{\partial \phi_1}{\partial x} \right|_{z=0} = -ik_x (A_1 + A') e^{i(\omega t - k_x x)} \quad (57)$$

$$\left. \frac{\partial \phi_1}{\partial z} \right|_{z=0} = ik_{z\alpha_1} (A_1 - A') e^{i(\omega t - k_x x)} \quad (58)$$

For $\psi_1 = B_1 e^{i(\omega t - k_x x - k_{z\beta_1} z)}$ at $z = 0$:

$$\left. \frac{\partial \psi_1}{\partial x} \right|_{z=0} = -ik_x B_1 e^{i(\omega t - k_x x)} \quad (59)$$

$$\left. \frac{\partial \psi_1}{\partial z} \right|_{z=0} = -ik_{z\beta_1} B_1 e^{i(\omega t - k_x x)} \quad (60)$$

Similarly for medium 2 at $z = 0$:

$$\phi_2 = A_2 e^{i(\omega t - k_x x + k_{z\alpha_2} z)} \quad (61)$$

$$\psi_2 = B_2 e^{i(\omega t - k_x x + k_{z\beta_2} z)} \quad (62)$$

$$\left. \frac{\partial \phi_2}{\partial x} \right|_{z=0} = -ik_x A_2 e^{i(\omega t - k_x x)} \quad (63)$$

$$\left. \frac{\partial \phi_2}{\partial z} \right|_{z=0} = ik_{z\alpha_2} A_2 e^{i(\omega t - k_x x)} \quad (64)$$

$$\left. \frac{\partial \psi_2}{\partial x} \right|_{z=0} = -ik_x B_2 e^{i(\omega t - k_x x)} \quad (65)$$

$$\left. \frac{\partial \psi_2}{\partial z} \right|_{z=0} = ik_{z\beta_2} B_2 e^{i(\omega t - k_x x)} \quad (66)$$

Boundary Condition 1: Horizontal Displacement

$$u_x^{(1)}|_{z=0} = u_x^{(2)}|_{z=0} \quad (67)$$

$$\left. \frac{\partial \phi_1}{\partial x} - \frac{\partial \psi_1}{\partial z} \right|_{z=0} = \left. \frac{\partial \phi_2}{\partial x} - \frac{\partial \psi_2}{\partial z} \right|_{z=0} \quad (68)$$

$$-ik_x(A_1 + A') - (-ik_{z\beta_1} B_1) = -ik_x A_2 - ik_{z\beta_2} B_2 \quad (69)$$

Simplifying:

$$k_x(A_1 + A') - k_{z\beta_1} B_1 = k_x A_2 + k_{z\beta_2} B_2 \quad (70)$$

Rearrange:

$$k_x A' - k_{z\beta_1} B_1 - k_x A_2 - k_{z\beta_2} B_2 = -k_x A_1$$

Boundary Condition 2: Vertical Displacement

$$u_z^{(1)}|_{z=0} = u_z^{(2)}|_{z=0} \quad (71)$$

$$\frac{\partial \phi_1}{\partial z} + \frac{\partial \psi_1}{\partial x} \Big|_{z=0} = \frac{\partial \phi_2}{\partial z} + \frac{\partial \psi_2}{\partial x} \Big|_{z=0} \quad (72)$$

$$ik_{z\alpha_1}(A_1 - A') + (-ik_x B_1) = ik_{z\alpha_2}A_2 + (-ik_x B_2) \quad (73)$$

Simplifying:

$$k_{z\alpha_1}(A_1 - A') - k_x B_1 = k_{z\alpha_2}A_2 - k_x B_2 \quad (74)$$

$$-k_{z\alpha_1}A' - k_x B_1 - k_{z\alpha_2}A_2 + k_x B_2 = -k_{z\alpha_1}A_1$$

Boundary Condition 3: Shear Stress

$$\sigma_{xz}^{(1)}|_{z=0} = \sigma_{xz}^{(2)}|_{z=0} \quad (75)$$

After computing second derivatives:

$$\mu_1[2\partial_{xz}\phi_1 + \partial_{xx}\psi_1 - \partial_{zz}\psi_1]|_{z=0} = \mu_2[2\partial_{xz}\phi_2 + \partial_{xx}\psi_2 - \partial_{zz}\psi_2]|_{z=0} \quad (76)$$

Computing the second and mixed derivatives using equations 57–66:

$$\partial_{xz}\phi_1|_{z=0} = (-ik_x)(ik_{z\alpha_1})(A_1 - A')e^{i(\omega t - k_x x)} = k_x k_{z\alpha_1}(A_1 - A')e^{i(\omega t - k_x x)} \quad (77)$$

$$\partial_{xx}\psi_1|_{z=0} = (-ik_x)^2 B_1 e^{i(\omega t - k_x x)} = -k_x^2 B_1 e^{i(\omega t - k_x x)} \quad (78)$$

$$\partial_{zz}\psi_1|_{z=0} = (-ik_{z\beta_1})^2 B_1 e^{i(\omega t - k_x x)} = -k_{z\beta_1}^2 B_1 e^{i(\omega t - k_x x)} \quad (79)$$

$$\partial_{xz}\phi_2|_{z=0} = (-ik_x)(ik_{z\alpha_2})A_2 e^{i(\omega t - k_x x)} = k_x k_{z\alpha_2}A_2 e^{i(\omega t - k_x x)} \quad (80)$$

$$\partial_{xx}\psi_2|_{z=0} = (-ik_x)^2 B_2 e^{i(\omega t - k_x x)} = -k_x^2 B_2 e^{i(\omega t - k_x x)} \quad (81)$$

$$\partial_{zz}\psi_2|_{z=0} = (ik_{z\beta_2})^2 B_2 e^{i(\omega t - k_x x)} = -k_{z\beta_2}^2 B_2 e^{i(\omega t - k_x x)} \quad (82)$$

Substitute the derivatives and cancel the common terms, this yields:

$$\mu_1[2k_x k_{z\alpha_1}(A_1 - A') + (k_{z\beta_1}^2 - k_x^2)B_1] = \mu_2[2k_x k_{z\alpha_2}A_2 + (k_{z\beta_2}^2 - k_x^2)B_2] \quad (83)$$

$$-2\mu_1 k_x k_{z\alpha_1} A' + \mu_1 (k_{z\beta_1}^2 - k_x^2) B_1 - 2\mu_2 k_x k_{z\alpha_2} A_2 - \mu_2 (k_{z\beta_2}^2 - k_x^2) B_2 = -2\mu_1 k_x k_{z\alpha_1} A_1$$

Multiply both sides by -1:

$$2\mu_1 k_x k_{z\alpha_1} A' + \mu_1 (k_x^2 - k_{z\beta_1}^2) B_1 + 2\mu_2 k_x k_{z\alpha_2} A_2 - \mu_2 (k_x^2 - k_{z\beta_2}^2) B_2 = 2\mu_1 k_x k_{z\alpha_1} A_1$$

Boundary Condition 4: Normal Stress

$$\sigma_{zz1}|_{z=0} = \sigma_{zz2}|_{z=0} \quad (84)$$

$$\begin{aligned} \lambda_1 (\partial_{xx} \phi_1 + \partial_{zz} \phi_1) + 2\mu_1 (\partial_{zz} \phi_1 + \partial_{xz} \psi_1) \Big|_{z=0} \\ = \lambda_2 (\partial_{xx} \phi_2 + \partial_{zz} \phi_2) + 2\mu_2 (\partial_{zz} \phi_2 + \partial_{xz} \psi_2) \Big|_{z=0} \end{aligned} \quad (85)$$

Computing:

$$\partial_{xx} \phi_1|_{z=0} = -k_x^2 (A_1 + A') e^{i(\omega t - k_x x)} \quad (86)$$

$$\partial_{zz} \phi_1|_{z=0} = -k_{z\alpha_1}^2 (A_1 + A') e^{i(\omega t - k_x x)} \quad (87)$$

$$\partial_{xz} \psi_1|_{z=0} = -k_x k_{z\beta_1} B_1 e^{i(\omega t - k_x x)} \quad (88)$$

$$\partial_{xx} \phi_2|_{z=0} = -k_x^2 A_2 e^{i(\omega t - k_x x)} \quad (89)$$

$$\partial_{zz} \phi_2|_{z=0} = -k_{z\alpha_2}^2 A_2 e^{i(\omega t - k_x x)} \quad (90)$$

$$\partial_{xz} \psi_2|_{z=0} = k_x k_{z\beta_2} B_2 e^{i(\omega t - k_x x)} \quad (91)$$

After substitution and simplification:

$$\begin{aligned} \lambda_1 (-k_x^2 - k_{z\alpha_1}^2) (A_1 + A') + 2\mu_1 (-k_{z\alpha_1}^2) (A_1 + A') - 2\mu_1 k_x k_{z\beta_1} B_1 \\ = \lambda_2 (-k_x^2 - k_{z\alpha_2}^2) A_2 + 2\mu_2 (-k_{z\alpha_2}^2) A_2 + 2\mu_2 k_x k_{z\beta_2} B_2 \end{aligned} \quad (92)$$

This can be rewritten as:

$$\begin{aligned}
& [\lambda_1(k_x^2 + k_{z\alpha_1}^2) + 2\mu_1 k_{z\alpha_1}^2] (A_1 + A') + 2\mu_1 k_x k_{z\beta_1} B_1 \\
& = [\lambda_2(k_x^2 + k_{z\alpha_2}^2) + 2\mu_2 k_{z\alpha_2}^2] A_2 - 2\mu_2 k_x k_{z\beta_2} B_2 \quad (93)
\end{aligned}$$

$$\begin{aligned}
& \left[\lambda_1(k_x^2 + k_{z\alpha_1}^2) + 2\mu_1 k_{z\alpha_1}^2 \right] A' + 2\mu_1 k_x k_{z\beta_1} B_1 - \left[\lambda_2(k_x^2 + k_{z\alpha_2}^2) + 2\mu_2 k_{z\alpha_2}^2 \right] A_2 + 2\mu_2 k_x k_{z\beta_2} B_2 \\
& = - \left[\lambda_1(k_x^2 + k_{z\alpha_1}^2) + 2\mu_1 k_{z\alpha_1}^2 \right] A_1 \quad (94)
\end{aligned}$$

Let:

$$M_1 = \lambda_1(k_x^2 + k_{z\alpha_1}^2) + 2\mu_1 k_{z\alpha_1}^2$$

$$M_2 = \lambda_2(k_x^2 + k_{z\alpha_2}^2) + 2\mu_2 k_{z\alpha_2}^2$$

Using $\lambda_m = \rho_m \alpha_m^2 - 2\mu_m$, $\mu_m = \rho_m \beta_m^2$ and $k_x^2 + k_{z\alpha_m}^2 = \omega^2 / \alpha_m^2$, these simplify to the useful closed form

$$M_m = \rho_m \omega^2 - 2\mu_m k_x^2 = \mu_m (k_{z\beta_m}^2 - k_x^2), \quad m = 1, 2.$$

$$M_1 A' + 2\mu_1 k_x k_{z\beta_1} B_1 - M_2 A_2 + 2\mu_2 k_x k_{z\beta_2} B_2 = -M_1 A_1$$

Matrix Form (Zoeppritz System)

Collecting the four equations from the boundary conditions into matrix form, $\mathbf{M}x = \mathbf{b}$, where $\mathbf{b}$ is the right-hand vector containing the incident-wave terms (proportional to A_1). Explicitly the coefficient matrix (rows are eqns from the BCs, columns unknowns $[A', B_1, A_2, B_2]$) is:

$$\begin{pmatrix}
k_x & -k_{z\beta_1} & -k_x & -k_{z\beta_2} \\
-k_{z\alpha_1} & -k_x & -k_{z\alpha_2} & k_x \\
2\mu_1 k_x k_{z\alpha_1} & \mu_1 (k_x^2 - k_{z\beta_1}^2) & 2\mu_2 k_x k_{z\alpha_2} & -\mu_2 (k_x^2 - k_{z\beta_2}^2) \\
M_1 & 2\mu_1 k_x k_{z\beta_1} & -M_2 & 2\mu_2 k_x k_{z\beta_2}
\end{pmatrix}
\begin{pmatrix}
A' \\
B_1 \\
A_2 \\
B_2
\end{pmatrix}
=
\begin{pmatrix}
-k_x A_1 \\
-k_{z\alpha_1} A_1 \\
2\mu_1 k_x k_{z\alpha_1} A_1 \\
-M_1 A_1
\end{pmatrix}$$

Divide the entire system by A_1 :

$$\begin{pmatrix} k_x & -k_{z\beta 1} & -k_x & -k_{z\beta 2} \\ -k_{z\alpha 1} & -k_x & -k_{z\alpha 2} & k_x \\ 2\mu_1 k_x k_{z\alpha 1} & \mu_1(k_x^2 - k_{z\beta 1}^2) & 2\mu_2 k_x k_{z\alpha 2} & -\mu_2(k_x^2 - k_{z\beta 2}^2) \\ M_1 & 2\mu_1 k_x k_{z\beta 1} & -M_2 & 2\mu_2 k_x k_{z\beta 2} \end{pmatrix} \begin{pmatrix} R_{PP} \\ R_{PS} \\ T_{PP} \\ T_{PS} \end{pmatrix} = \begin{pmatrix} -k_x \\ -k_{z\alpha 1} \\ 2\mu_1 k_x k_{z\alpha 1} \\ -M_1 \end{pmatrix}$$

Where:

$$M_1 = \lambda_1(k_x^2 + k_{z\alpha 1}^2) + 2\mu_1 k_{z\alpha 1}^2$$

$$M_2 = \lambda_2(k_x^2 + k_{z\alpha 2}^2) + 2\mu_2 k_{z\alpha 2}^2$$

The reflection and transmission coefficients are therefore:

$$R_{PP} = \frac{A'}{A_1}, \quad R_{PS} = \frac{B_1}{A_1}, \quad T_{PP} = \frac{A_2}{A_1}, \quad T_{PS} = \frac{B_2}{A_1}.$$

This 4×4 linear system is the general **Zoeppritz equation** for a P-wave incident on a plane elastic interface. Solving for RC and TC gives:

$$\begin{pmatrix} R_{PP} \\ R_{PS} \\ T_{PP} \\ T_{PS} \end{pmatrix} = \mathbf{M}^{-1} \mathbf{b}.$$

Where:

$$\mathbf{M} = \begin{pmatrix} k_x & -k_{z\beta 1} & -k_x & -k_{z\beta 2} \\ -k_{z\alpha 1} & -k_x & -k_{z\alpha 2} & k_x \\ 2\mu_1 k_x k_{z\alpha 1} & \mu_1(k_x^2 - k_{z\beta 1}^2) & 2\mu_2 k_x k_{z\alpha 2} & -\mu_2(k_x^2 - k_{z\beta 2}^2) \\ M_1 & 2\mu_1 k_x k_{z\beta 1} & -M_2 & 2\mu_2 k_x k_{z\beta 2} \end{pmatrix}$$

$$\mathbf{b} = \begin{pmatrix} -k_x \\ -k_{z\alpha 1} \\ 2\mu_1 k_x k_{z\alpha 1} \\ -M_1 \end{pmatrix}$$

The Zoeppritz system of equations above, in its potential-based form, can be transformed into the more compact form presented in the classical book by Aki and Richards. Note

that the coefficients defined here are ratios of *potential* amplitudes. Since the displacement amplitude of a plane wave is (ω/v) times its potential amplitude, the displacement-amplitude coefficients tabulated by Aki and Richards differ by velocity-ratio factors, e.g. $R_{PS}^{\text{disp}} = (\alpha_1/\beta_1) R_{PS}$ and $T_{PP}^{\text{disp}} = (\alpha_1/\alpha_2) T_{PP}$. This transformation is carried out explicitly below.

Generalized Trigonometric Form (Angles, V_P , V_S)

Three steps convert the wavenumber-domain matrix into the classical angle-domain form.

Step 1: Snell substitutions. Every wavenumber is ω times a slowness:

$$k_x = \frac{\omega \sin i_1}{\alpha_1} = \frac{\omega \sin j_1}{\beta_1} = \frac{\omega \sin i_2}{\alpha_2} = \frac{\omega \sin j_2}{\beta_2}, \quad k_{z\alpha_m} = \frac{\omega \cos i_m}{\alpha_m}, \quad k_{z\beta_m} = \frac{\omega \cos j_m}{\beta_m}$$

Crucially, Snell's law lets us express k_x in each *column* of the matrix through the angle of that column's own wave (e.g. $k_x = \omega \sin j_2/\beta_2$ in the T_{PS} column).

Step 2: Trigonometric identities. Using $\mu = \rho\beta^2$ and the double-angle identities, each matrix entry collapses (per medium, with i the P angle and j the SV angle):

$$2\mu k_x k_{z\alpha} = 2\rho\beta^2 \frac{\omega^2}{\alpha^2} \sin i \cos i = \rho\omega^2 \frac{\beta^2}{\alpha^2} \sin 2i \quad (95)$$

$$2\mu k_x k_{z\beta} = 2\rho\beta^2 \frac{\omega^2}{\beta^2} \sin j \cos j = \rho\omega^2 \sin 2j \quad (96)$$

$$\mu (k_x^2 - k_{z\beta}^2) = \rho\omega^2 (\sin^2 j - \cos^2 j) = -\rho\omega^2 \cos 2j \quad (97)$$

$$M = \mu (k_{z\beta}^2 - k_x^2) = \rho\omega^2 \cos 2j \quad (98)$$

where the last line uses the closed form $M_m = \mu_m(k_{z\beta_m}^2 - k_x^2)$ derived above.

Step 3: Rescale to displacement amplitudes. Substituting

$$R_{PP} = \tilde{R}_{PP}, \quad R_{PS} = \frac{\beta_1}{\alpha_1} \tilde{R}_{PS}, \quad T_{PP} = \frac{\alpha_2}{\alpha_1} \tilde{T}_{PP}, \quad T_{PS} = \frac{\beta_2}{\alpha_1} \tilde{T}_{PS}$$

and then multiplying rows 1–2 by α_1/ω and rows 3–4 by α_1/ω^2 , every power of ω cancels — confirming that the R/T coefficients are frequency-independent. The result is the **generalized trigonometric Zoeppritz system**:

$$\begin{pmatrix} \sin i_1 & -\cos j_1 & -\sin i_2 & -\cos j_2 \\ -\cos i_1 & -\sin j_1 & -\cos i_2 & \sin j_2 \\ \rho_1 \frac{\beta_1^2}{\alpha_1} \sin 2i_1 & -\rho_1 \beta_1 \cos 2j_1 & \rho_2 \frac{\beta_2^2}{\alpha_2} \sin 2i_2 & \rho_2 \beta_2 \cos 2j_2 \\ \rho_1 \alpha_1 \cos 2j_1 & \rho_1 \beta_1 \sin 2j_1 & -\rho_2 \alpha_2 \cos 2j_2 & \rho_2 \beta_2 \sin 2j_2 \end{pmatrix} \begin{pmatrix} \tilde{R}_{PP} \\ \tilde{R}_{PS} \\ \tilde{T}_{PP} \\ \tilde{T}_{PS} \end{pmatrix} = \begin{pmatrix} -\sin i_1 \\ -\cos i_1 \\ \rho_1 \frac{\beta_1^2}{\alpha_1} \sin 2i_1 \\ -\rho_1 \alpha_1 \cos 2j_1 \end{pmatrix}$$

The four rows are, in order: continuity of u_x , of u_z , of σ_{xz} , and of σ_{zz} . This is the classical Zoeppritz system in terms of angles, V_P (α), V_S (β), and densities — up to the sign conventions for the positive displacement directions of each wave, which differ among textbooks. Beyond a critical angle the corresponding cosine is continued analytically, $\cos \theta = \sqrt{1 - p^2 v^2}$ becoming imaginary, and the coefficients become complex.

Consistency check: normal incidence. At $i_1 = j_1 = i_2 = j_2 = 0$ the system decouples:

- Row 1: $-\tilde{R}_{PS} - \tilde{T}_{PS} = 0$
- Row 3: $-\rho_1 \beta_1 \tilde{R}_{PS} + \rho_2 \beta_2 \tilde{T}_{PS} = 0 \Rightarrow \tilde{R}_{PS} = \tilde{T}_{PS} = 0$ (no mode conversion)
- Row 2: $\tilde{R}_{PP} + \tilde{T}_{PP} = 1$
- Row 4: $\rho_1 \alpha_1 \tilde{R}_{PP} - \rho_2 \alpha_2 \tilde{T}_{PP} = -\rho_1 \alpha_1$

Solving the last two equations with $Z_m = \rho_m \alpha_m$:

$$\tilde{R}_{PP} \Big|_{i_1=0} = \frac{Z_2 - Z_1}{Z_2 + Z_1}, \quad \tilde{T}_{PP} \Big|_{i_1=0} = \frac{2Z_1}{Z_1 + Z_2}$$

which are the familiar normal-incidence impedance formulas — an independent check that the trigonometric system (and hence the wavenumber-domain matrix it came from) is correct.

7. Special Case: P-SV R/T Coefficients at Free Surface (Earth's Surface)

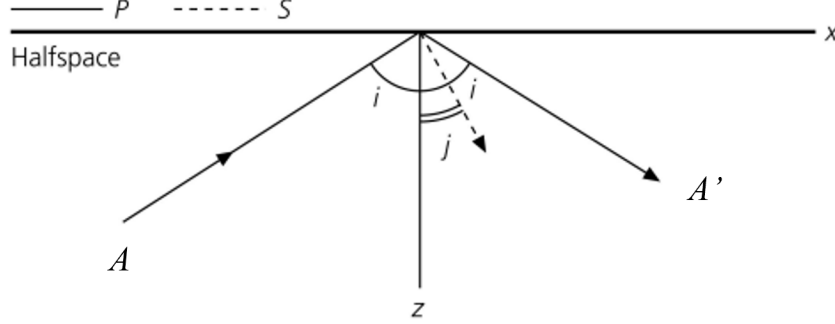


Figure 2: P-SV R/T Coefficients at Earth Surface

The free surface (e.g., the Earth's surface) is a special case where Medium 2 is a vacuum. This simplifies the problem because there are no transmitted waves and the boundary conditions become **stress-free**.

At Earth's surface, the boundary conditions simplify to **stress-free conditions**:

$$\sigma_{xz}|_{z=0} = 0, \quad \sigma_{zz}|_{z=0} = 0 \quad (99)$$

This gives us two equations for two unknowns (A' , B_1):

Free Surface Boundary Conditions

- Medium 1: Elastic solid with parameters $\alpha, \beta, \rho, \lambda, \mu$
- Medium 2: Vacuum ($z < 0$, the transmission side in the sign convention of Section 2). As drawn in Figure 2, z is taken positive downward: the solid halfspace occupies $z > 0$, the vacuum lies above the surface, and the incident P-wave A_1 travels upward (the $-z$ direction) toward the surface.
- Boundary conditions at $z = 0$: $\sigma_{zz} = 0$ and $\sigma_{xz} = 0$

Shear Stress = 0

From Boundary Condition 3 with medium 2 absent:

$$\mu[2k_x k_{z\alpha}(A_1 - A') + (k_{z\beta}^2 - k_x^2)B_1] = 0 \quad (100)$$

$$2k_x k_{z\alpha}(A_1 - A') + (k_{z\beta}^2 - k_x^2)B_1 = 0 \quad (101)$$

Dividing by k_x^2 :

$$2\frac{k_{z\alpha}}{k_x}(A_1 - A') + \left(\frac{k_{z\beta}^2}{k_x^2} - 1\right) B_1 = 0 \quad (102)$$

Defining:

$$\boxed{r_\alpha = \frac{k_{z\alpha}}{k_x} = \cot i_\alpha \quad \text{and} \quad r_\beta = \frac{k_{z\beta}}{k_x} = \cot i_\beta}$$

Note that $\frac{k_{z\beta}^2}{k_x^2} = r_\beta^2$, so:

$$\boxed{2r_\alpha(A_1 - A') + (r_\beta^2 - 1)B_1 = 0} \quad (103)$$

Normal Stress = 0

From Boundary Condition 4 with medium 2 absent:

$$\lambda(k_x^2 + k_{z\alpha}^2)(A_1 + A') + 2\mu(k_{z\alpha}^2)(A_1 + A') + 2\mu k_x k_{z\beta} B_1 = 0$$

$$[\lambda(k_x^2 + k_{z\alpha}^2) + 2\mu k_{z\alpha}^2](A_1 + A') + 2\mu k_x k_{z\beta} B_1 = 0 \quad (104)$$

We can rewrite:

$$[\lambda(k_x^2 + k_{z\alpha}^2) + 2\mu k_{z\alpha}^2](A_1 + A') = -2\mu k_x k_{z\beta} B_1 \quad (105)$$

Dividing by k_x^2 :

$$\left[\lambda\left(1 + \frac{k_{z\alpha}^2}{k_x^2}\right) + 2\mu\frac{k_{z\alpha}^2}{k_x^2}\right](A_1 + A') = -2\mu\frac{k_{z\beta}}{k_x} B_1 \quad (106)$$

$$[\lambda(1 + r_\alpha^2) + 2\mu r_\alpha^2](A_1 + A') = -2\mu r_\beta B_1 \quad (107)$$

This can be rewritten as:

$$\lambda(1 + r_\alpha^2)(A_1 + A') + 2\mu r_\alpha^2(A_1 + A') + 2\mu r_\beta B_1 = 0$$

$$\boxed{\lambda(1 + r_\alpha^2)(A_1 + A') + 2\mu(r_\alpha^2(A_1 + A') + r_\beta B_1) = 0} \quad (108)$$

Solving for Reflection Coefficients

System of Equations

From Equation 103:

$$2r_\alpha(A_1 - A') + (r_\beta^2 - 1)B_1 = 0$$

$$\boxed{A_1 - A' = -\frac{(r_\beta^2 - 1)}{2r_\alpha}B_1 = \frac{(1 - r_\beta^2)}{2r_\alpha}B_1} \quad (109)$$

From Equation 108:

$$\lambda(1 + r_\alpha^2)(A_1 + A') + 2\mu(r_\alpha^2(A_1 + A') + r_\beta B_1) = 0$$

$$[\lambda(1 + r_\alpha^2) + 2\mu r_\alpha^2] (A_1 + A') = -2\mu r_\beta B_1$$

$$\boxed{A_1 + A' = \frac{-2\mu r_\beta B_1}{\lambda(1 + r_\alpha^2) + 2\mu r_\alpha^2}} \quad (110)$$

Solving by Elimination Method

Let:

$$S = A_1 + A' \quad (\text{sum}) \quad (111)$$

$$D = A_1 - A' \quad (\text{difference}) \quad (112)$$

Then: $A_1 = (S + D)/2$ and $A' = (S - D)/2$

From Eq. 109:

$$D = \frac{(1 - r_\beta^2)}{2r_\alpha}B_1 \quad (113)$$

From Eq. 110:

$$S = -\frac{2\mu r_\beta}{\lambda(1 + r_\alpha^2) + 2\mu r_\alpha^2}B_1 \quad (114)$$

Therefore:

$$A_1 = \frac{1}{2} \left[\frac{(1 - r_\beta^2)}{2r_\alpha} - \frac{2\mu r_\beta}{\lambda(1 + r_\alpha^2) + 2\mu r_\alpha^2} \right] B_1 \quad (115)$$

$$A' = \frac{1}{2} \left[-\frac{2\mu r_\beta}{\lambda(1 + r_\alpha^2) + 2\mu r_\alpha^2} - \frac{(1 - r_\beta^2)}{2r_\alpha} \right] B_1 \quad (116)$$

Key Identity

The denominator combination $\lambda(1 + r_\alpha^2) + 2\mu r_\alpha^2$ simplifies to a single term. Multiplying by k_x^2 and using $k_x^2 + k_{z\alpha}^2 = k_\alpha^2 = \omega^2/\alpha^2$, $\lambda = \rho\alpha^2 - 2\mu$, and $\mu = \rho\beta^2$ (so that $\rho\omega^2 = \mu k_\beta^2$ with $k_\beta^2 = k_x^2 + k_{z\beta}^2$):

$$\begin{aligned}\lambda k_\alpha^2 + 2\mu k_{z\alpha}^2 &= \rho\omega^2 - 2\mu k_\alpha^2 + 2\mu k_{z\alpha}^2 = \rho\omega^2 - 2\mu k_x^2 \\ &= \mu(k_\beta^2 - 2k_x^2) = \mu(k_{z\beta}^2 - k_x^2)\end{aligned}\tag{117}$$

Dividing by k_x^2 :

$$\boxed{\lambda(1 + r_\alpha^2) + 2\mu r_\alpha^2 = \mu(r_\beta^2 - 1)}\tag{118}$$

so that

$$S = -\frac{2r_\beta}{r_\beta^2 - 1}B_1\tag{119}$$

Final Expressions

Since $A_1 = (S + D)/2$ and $A' = (S - D)/2$, the coefficients are $R_P = A'/A_1 = (S - D)/(S + D)$ and $R_{SV} = B_1/A_1 = 2B_1/(S + D)$. Explicitly, putting S and D over the common denominator $2r_\alpha(r_\beta^2 - 1)$:

$$S + D = -\left[\frac{2r_\beta}{r_\beta^2 - 1} + \frac{r_\beta^2 - 1}{2r_\alpha}\right]B_1 = -\frac{4r_\alpha r_\beta + (r_\beta^2 - 1)^2}{2r_\alpha(r_\beta^2 - 1)}B_1\tag{120}$$

$$S - D = -\left[\frac{2r_\beta}{r_\beta^2 - 1} - \frac{r_\beta^2 - 1}{2r_\alpha}\right]B_1 = -\frac{4r_\alpha r_\beta - (r_\beta^2 - 1)^2}{2r_\alpha(r_\beta^2 - 1)}B_1\tag{121}$$

In the ratios, the common factor $-B_1/[2r_\alpha(r_\beta^2 - 1)]$ cancels, and the reflection coefficients follow:

P-to-P Reflection Coefficient:

$$\boxed{R_P = \frac{A'}{A_1} = \frac{4r_\alpha r_\beta - (r_\beta^2 - 1)^2}{4r_\alpha r_\beta + (r_\beta^2 - 1)^2}}\tag{122}$$

P-to-SV Conversion Coefficient:

$$R_{SV} = \frac{B_1}{A_1} = \frac{4r_\alpha(1 - r_\beta^2)}{4r_\alpha r_\beta + (r_\beta^2 - 1)^2} \quad (123)$$

where:

$$r_\alpha = \frac{k_{z\alpha}}{k_x} = \cot i_\alpha = \frac{\cos i_\alpha}{\sin i_\alpha} \quad (124)$$

$$r_\beta = \frac{k_{z\beta}}{k_x} = \cot i_\beta = \frac{\cos i_\beta}{\sin i_\beta} \quad (125)$$

Note: The angles i_α and i_β are measured from the vertical (normal to the surface); they are the angles labeled i and j in Figure 2.

Remember the relations:

- $\lambda + 2\mu = \rho\alpha^2$
- $\mu = \rho\beta^2$
- $k_x^2 + k_{z\alpha}^2 = \omega^2/\alpha^2 = k_\alpha^2$

8. Physical Interpretation

Key Features

1. **Mode Conversion:** Unlike SH waves, P-waves generate both P and SV reflected waves at a free surface.
2. **No Critical Angle for Incident P:** Because $\beta < \alpha$, Snell's law gives $\sin i_\beta = (\beta/\alpha) \sin i_\alpha < 1$ for every incidence angle, so the reflected SV wave is always homogeneous. A critical angle at the free surface exists only for an *incident SV* wave: beyond $j_c = \sin^{-1}(\beta/\alpha)$ the reflected P wave becomes inhomogeneous (evanescent).
3. **Amplitude Behavior:**
 - At normal incidence ($i = 0$): $r_\alpha \rightarrow \infty$, $r_\beta \rightarrow \infty$, giving $R_P = -1$, $R_{SV} = 0$
 - At grazing incidence ($i \rightarrow 90$): $r_\alpha \rightarrow 0$ while $r_\beta \rightarrow \sqrt{\alpha^2/\beta^2 - 1}$ (finite, since $\sin i_\beta \rightarrow \beta/\alpha < 1$), giving $R_P \rightarrow -1$, $R_{SV} \rightarrow 0$
 - Maximum P-to-SV conversion occurs at intermediate angles
4. **Energy Conservation:** since R_P and R_{SV} are *potential*-amplitude ratios and displacement amplitudes scale as $(\omega/v) \times (\text{potential amplitude})$, the vertical energy-flux balance reads

$$|R_P|^2 + |R_{SV}|^2 \frac{\alpha \cos i_\beta}{\beta \cos i_\alpha} = 1 \quad (126)$$

For displacement-amplitude coefficients ($\tilde{R}_{SV} = (\alpha/\beta)R_{SV}$) the factor becomes $\frac{I_\beta \cos i_\beta}{I_\alpha \cos i_\alpha}$, where $I = \rho v$ is the impedance ($I_\alpha = \rho\alpha$, $I_\beta = \rho\beta$).

Special Values

At normal incidence ($i_\alpha = 0$, i.e., $k_x \rightarrow 0$ so $r_\alpha, r_\beta \rightarrow \infty$ with $r_\alpha/r_\beta \rightarrow \beta/\alpha$; the $(r_\beta^2 - 1)^2$ term dominates), taking the limit of the closed-form expressions:

$$R_P(0) = -1 \quad (\text{phase reversal}) \quad (127)$$

$$R_{SV}(0) = 0 \quad (\text{no mode conversion}) \quad (128)$$

There is no critical angle for an incident P wave at the free surface: since $\beta < \alpha$, $\sin i_\beta = (\beta/\alpha) \sin i_\alpha < 1$ for all i_α , so the reflected SV wave never grazes the surface and r_β never vanishes (its minimum value is $\sqrt{\alpha^2/\beta^2 - 1}$, reached at grazing P incidence). Mode conversion instead peaks smoothly at intermediate incidence angles (roughly 40–50 for typical α/β ratios).

9. Shuey's Approximation (Linearized AVO)

For small contrasts across the interface ($|\Delta\alpha|/\alpha, |\Delta\beta|/\beta, |\Delta\rho|/\rho \ll 1$) and pre-critical angles, linearizing the exact Zoeppritz solution for R_{PP} gives the **Aki–Richards approximation**:

$$R(\theta) \approx \frac{1}{2} (1 - 4p^2\beta^2) \frac{\Delta\rho}{\rho} + \frac{1}{2\cos^2\theta} \frac{\Delta\alpha}{\alpha} - 4p^2\beta^2 \frac{\Delta\beta}{\beta}, \quad p = \frac{\sin\theta}{\alpha} \quad (129)$$

Substituting $4p^2\beta^2 = 4\frac{\beta^2}{\alpha^2} \sin^2\theta$ and

$$\frac{1}{2\cos^2\theta} = \frac{1}{2} (1 + \tan^2\theta) = \frac{1}{2} + \frac{1}{2} \sin^2\theta + \frac{1}{2} (\tan^2\theta - \sin^2\theta),$$

then collecting the terms in 1, $\sin^2\theta$, and $(\tan^2\theta - \sin^2\theta)$, yields **Shuey's three-term form**:

$$\boxed{R(\theta) = R(0) + G \sin^2\theta + F(\tan^2\theta - \sin^2\theta)} \quad (130)$$

where:

Intercept (normal incidence reflectivity):

$$R(0) = \frac{1}{2} \left(\frac{\Delta\rho}{\rho} + \frac{\Delta\alpha}{\alpha} \right) \quad (131)$$

Gradient (AVO slope):

$$G = \frac{\Delta\alpha}{2\alpha} - \frac{2\beta^2}{\alpha^2} \left(\frac{\Delta\rho}{\rho} + \frac{2\Delta\beta}{\beta} \right) \quad (132)$$

Large-angle term:

$$F = \frac{\Delta\alpha}{2\alpha} \quad (133)$$

Here:

- $\Delta\rho = \rho_2 - \rho_1$, $\Delta\alpha = \alpha_2 - \alpha_1$, $\Delta\beta = \beta_2 - \beta_1$
- $\rho = (\rho_1 + \rho_2)/2$, $\alpha = (\alpha_1 + \alpha_2)/2$, $\beta = (\beta_1 + \beta_2)/2$
- θ is the angle of incidence (more precisely, the average of the incidence and transmission angles)

This approximation is widely used in seismic exploration for amplitude-versus-offset (AVO) analysis to detect hydrocarbons and characterize lithology.

Physical Meaning of AVO Parameters

- $R(0)$: Measures the impedance contrast at normal incidence
- G : Captures the angle-dependent behavior, sensitive to fluid content and Poisson's ratio changes
- F : Important at large angles, related to P-wave velocity contrast

Summary

The P-SV problem is significantly more complex than the SH problem due to:

1. Mode conversion between P and SV waves
2. Coupling through both displacement and stress boundary conditions
3. Four coupled equations instead of two
4. Angle-dependent behavior that depends on both α and β

The exact Zoeppritz equations provide the complete solution, while Shuey's approximation offers a practical, linearized alternative for seismic interpretation.